3. Data and Methods

3.1 Data Sources

Developing, training, and evaluating the proposed intelligent system for job matching and resume tailoring requires high-quality textual data. Three open-source datasets were selected from Hugging Face for this study. No web scraping or paid APIs were used to collect the data. The datasets contain CVs and job descriptions, which are used to develop and test the proposed system.

- **Resume dataset:** an extensive collection of data comprising over 2,400 resumes across 23 different professional categories. It serves as a primary database of candidate profiles.
- **Data Science job descriptions:** a collection comprising approximately 3,000 detailed job postings from the IT and Data Science sectors. This dataset provides the essential textual content - candidate requirements and job responsibilities that are needed for effective semantic indexing and job matching.
- **Another dataset** contains around 960 resumes paired with job descriptions. It is used to check whether an LLM can change a resume to better match a job while keeping the candidate's original information.

3.2 Data Formats and Storage

To keep the project organized and make the results easy to reproduce, a clear folder structure and file naming system were created. Raw data is stored separately from processed data.

The original datasets are stored in `data/raw/` in formats such as .csv and .jsonl. These files are not changed manually. All data processing is done using Python scripts in the `scripts/` directory.

The data processed is saved in `data/processed/`. File names have version numbers, like `resumes_cleaned_v1.csv`, to show which version is used and keep files from older versions. The `data/docs/` folder contains a `README.md` file and a data dictionary that describes the datasets, their columns, and data types.

3.3 Preprocessing and Cleaning

Since the system uses Natural Language Processing (NLP), Sentence-BERT embeddings, and Retrieval-Augmented Generation (RAG), the raw text data needed to be cleaned before use.

The preprocessing was done in Python using libraries such as pandas.

First, missing and incorrect data was removed. Rows with empty text or missing values were deleted. Very short texts (less than 50 words) were also removed because they do not contain enough information for accurate matching.

Second, the data was filtered to focus on the IT sector. The original Resume Dataset contained 23 different job categories, including both technical and non-technical roles. Only resumes with relevant categories, such as "INFORMATION-TECHNOLOGY", "DATA-SCIENCE", and "ENGINEERING"were kept.

Before processing, the text was prepared for each task. For semantic matching, everything was converted to lowercase, and HTML tags and special characters were removed. This helped prepare the text for creating embeddings. However, for the CV personalization module, the original capitalization, punctuation, and formatting were kept. This helps the RAG system generate clear and grammatically correct text.

3.4 Annotations and Labels

The system uses cosine similarity to compare CVs and job descriptions. Although it can do this without labeled data, examples with known results are needed to check how well the system works.

Reviewers manually score a subset of matches to evaluate actual job fit. This human baseline serves as ground truth to benchmark system performance and determine the optimal cut-off thresholds for rejecting poor matches.

The CV adaptation module also needs to be tested to make sure the generated CVs are suitable for professional use. A sample of adapted CVs will be reviewed and scored by human reviewers based on three criteria:

- **Factual Accuracy:** Ensuring the model does not hallucinate or add fake skills.
- **Alignment:** Highlighting the specific background that matches the target vacancy.
- **Readability:** Checking for natural phrasing and proper grammar.

3.5 Ethical and Legal Considerations

The AI tools used for recruiting need to stick to all privacy laws and ethical standards. Datasets that were used in this study are publicly available and are licensed under Apache-2.0, MIT, and

WTFPL licenses, allowing to use the data for research purposes according to the conditions outlined by each license.

All personal data including names, emails, phone numbers will be deleted.

The system will help users find suitable job offers and improve their resumes. However, the tool will not be used for making hiring decisions. Considering that LLMs can show some bias, the final decision will be up to users and recruiters. The users will also get the chance to check the offered job opportunities and resume before using them.

3.6 Presentation of Results

The results will be presented in tables and figures. Tables will show numerical results, such as similarity scores, system response time, and recommendation accuracy. To show how the RAG pipeline and language model work, examples of CVs before and after changes will be compared. Figures will be used to show how the system works, how the data is distributed, and how different scoring thresholds affect the results. Each table and figure will have a number and a short title. They will also be mentioned in the text to help readers understand the results.